Ex. No: 06	INTEGRATE ANIMATIONS AND TRANSITION
Date of Exercise	08.09.2026
Git-hub Host Link	https://fr8st7.github.io/flutter_exp_6/

Aim:

To Implement animations and smooth transitions in a Flutter application using AnimatedContainer.

Description:

This experiment demonstrates the use of Flutter's **AnimatedContainer** to create a smooth size transition. A button is provided to change the size of the widget dynamically using setState(). A simple **light-purple color theme** is used for the user interface.

Procedure:

Step 1: Create a new Flutter project using VS Code or the terminal.

flutter create animation_app

cd animation_app

code .

Step 2: Open the lib/main.dart file and remove the default Flutter code.

Step 3: Create a StatefulWidget to manage the animation state.

Step 4: Create a Boolean state variable isLarge to control the size of the animated widget.

Step 5: Add an AnimatedContainer with a duration of one second and set its width and height based on the isLarge value.

Step 6: Add an **Animate** button and use setState() to change the value of isLarge.

Step 7: Apply a light purple background, app bar, animated container, and button to create a simple user interface.

Step 8: Save the program and run the application using:

flutter run -d chrome

Step 9: Click the **Animate** button and observe the smooth change in the size of the widget.

Program:

```
import 'package:flutter/material.dart';
void main() {
  runApp(const AnimationApp());
}
class AnimationApp extends StatelessWidget {
  const AnimationApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
    debugShowCheckedModeBanner: false,
    title: 'Flutter Animation',
      theme: ThemeData(
        colorScheme: ColorScheme.fromSeed(
          seedColor: Colors.deepPurple,
        ),
        useMaterial3: true,
      ),
      home: const AnimationPage(),
    );
  }
}
class AnimationPage extends StatefulWidget {
  const AnimationPage({super.key});

  @override
  State<AnimationPage> createState() => _AnimationPageState();
}
class _AnimationPageState extends State<AnimationPage> {
  bool isLarge = false;
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text(
          'Flutter Animations',
          style: TextStyle(
          color: Colors.white,
            fontWeight: FontWeight.bold,
          ),
        ),
```

```
  centerTitle: true,
  backgroundColor: Colors.deepPurple,
),
body: Container(
  width: double.infinity,
  height: double.infinity,
  color: const Color(0xFFEDE7F6),

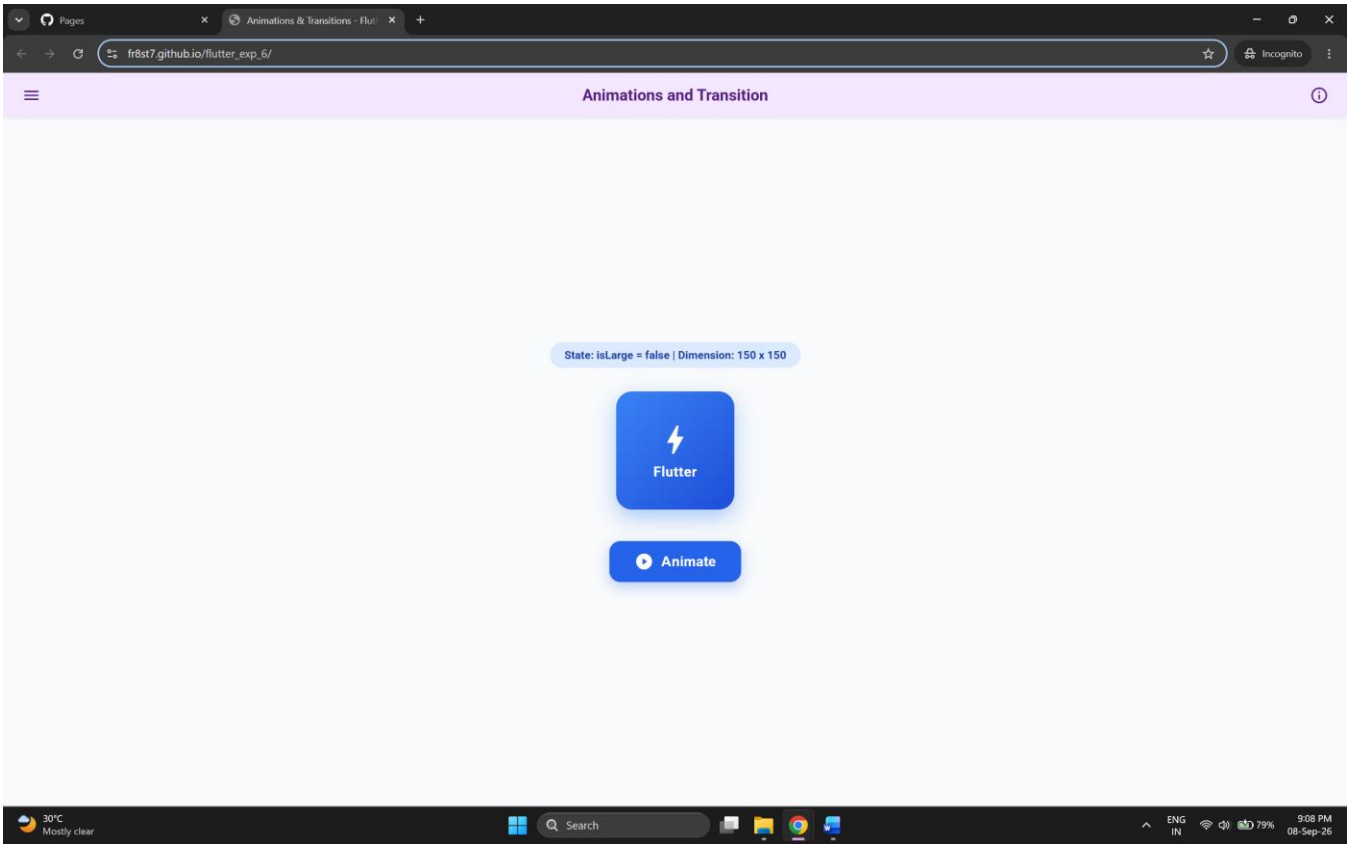
  child: Center(
    child: Column(
      mainAxisAlignment: MainAxisAlignment.center,
      children: [
        AnimatedContainer(
          duration: const Duration(seconds: 1),
          curve: Curves.easeInOut,
          width: isLarge ? 300 : 150,
          height: isLarge ? 300 : 150,
          decoration: BoxDecoration(
            color: const Color(0xFFB39DDB),
            borderRadius: BorderRadius.circular(
            isLarge ? 50 : 20,
            ),
            boxShadow: const [
              BoxShadow(
              blurRadius: 15,
                spreadRadius: 3,
                offset: Offset(0, 8),
              ),
            ],
          ),
          child: const Center(
            child: Text(
            'Flutter',
              style: TextStyle(
                fontSize: 28,
                color: Colors.white,
                fontWeight: FontWeight.bold,
              ),
            ),
          ),
        ),
        const SizedBox(height: 40),
```

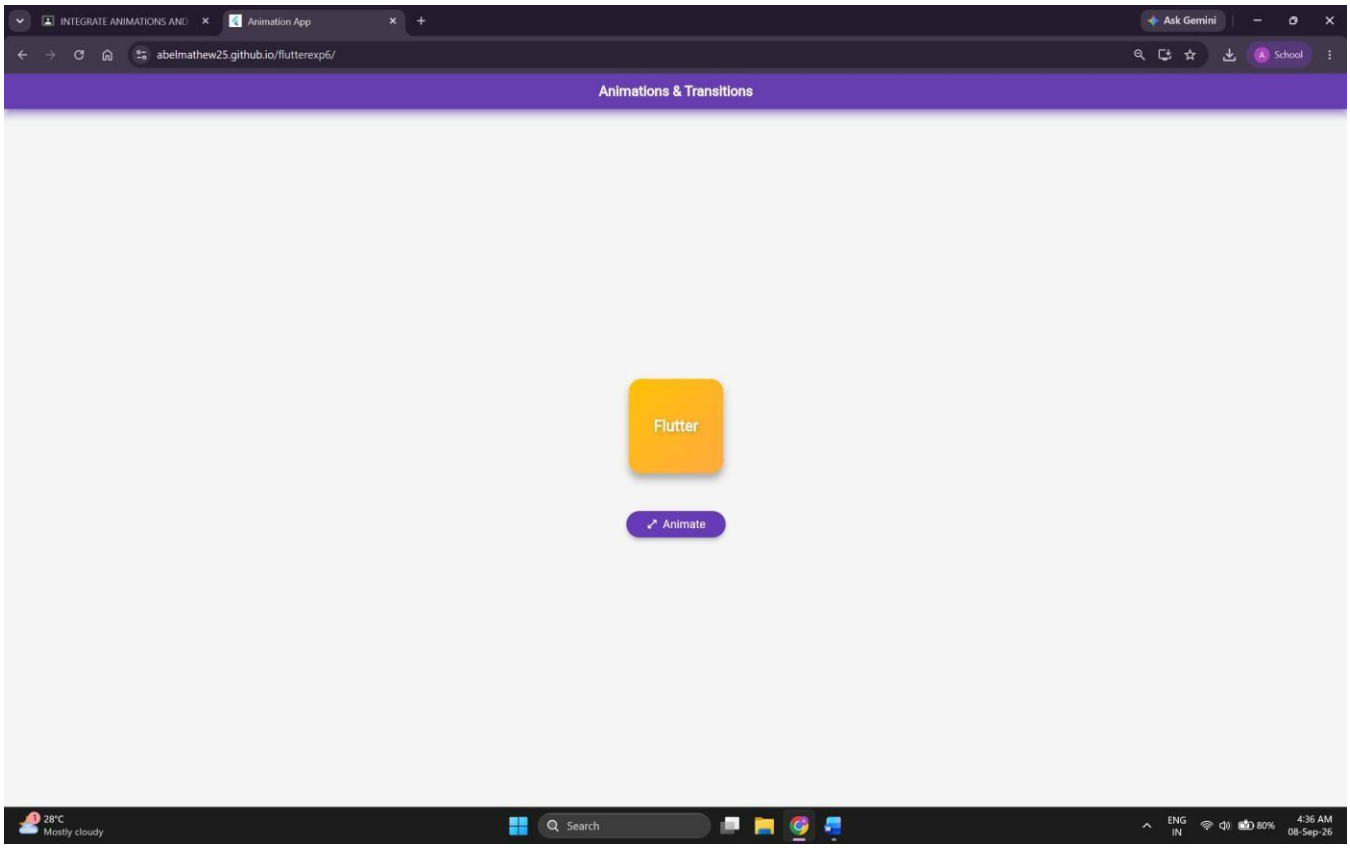
```
ElevatedButton.icon(
onPressed: () {
   setState(() {
     isLarge = !isLarge;
   });
 },
 icon: const Icon(Icons.animation),
 label: const Text(
   'Animate',
   style: TextStyle(
     fontSize: 18,
     fontWeight: FontWeight.bold,
   ),
 ),
 style: ElevatedButton.styleFrom(
   backgroundColor: Colors.deepPurple,
   foregroundColor: Colors.white,
   padding: const EdgeInsets.symmetric(
   horizontal: 35,
     vertical: 16,
   ),
   shape: RoundedRectangleBorder(
     borderRadius: BorderRadius.circular(30),
   ),
 ),
),

const SizedBox(height: 20),

Text(
 isLarge ? 'Widget is Large' : 'Widget is Small',
 style: const TextStyle(
   color: Colors.deepPurple,
   fontSize: 18,
   fontWeight: FontWeight.w500,
    ),),],),),),);}}
```

Output:


Pages
Animations & Transitions - Flut
fr8st7.github.io/flutter_exp_6/
Incognito
Animations and Transition
State: isLarge = false | Dimension: 150 x 150
Flutter
Animate
30°C
Mostly clear
Search
ENG
IN
79%
9:08 PM
08-Sep-26



Result:

The Flutter application was successfully implemented using **AnimatedContainer**. The widget smoothly changes its size when the **Animate** button is clicked, demonstrating basic animation and transition functionality in Flutter.